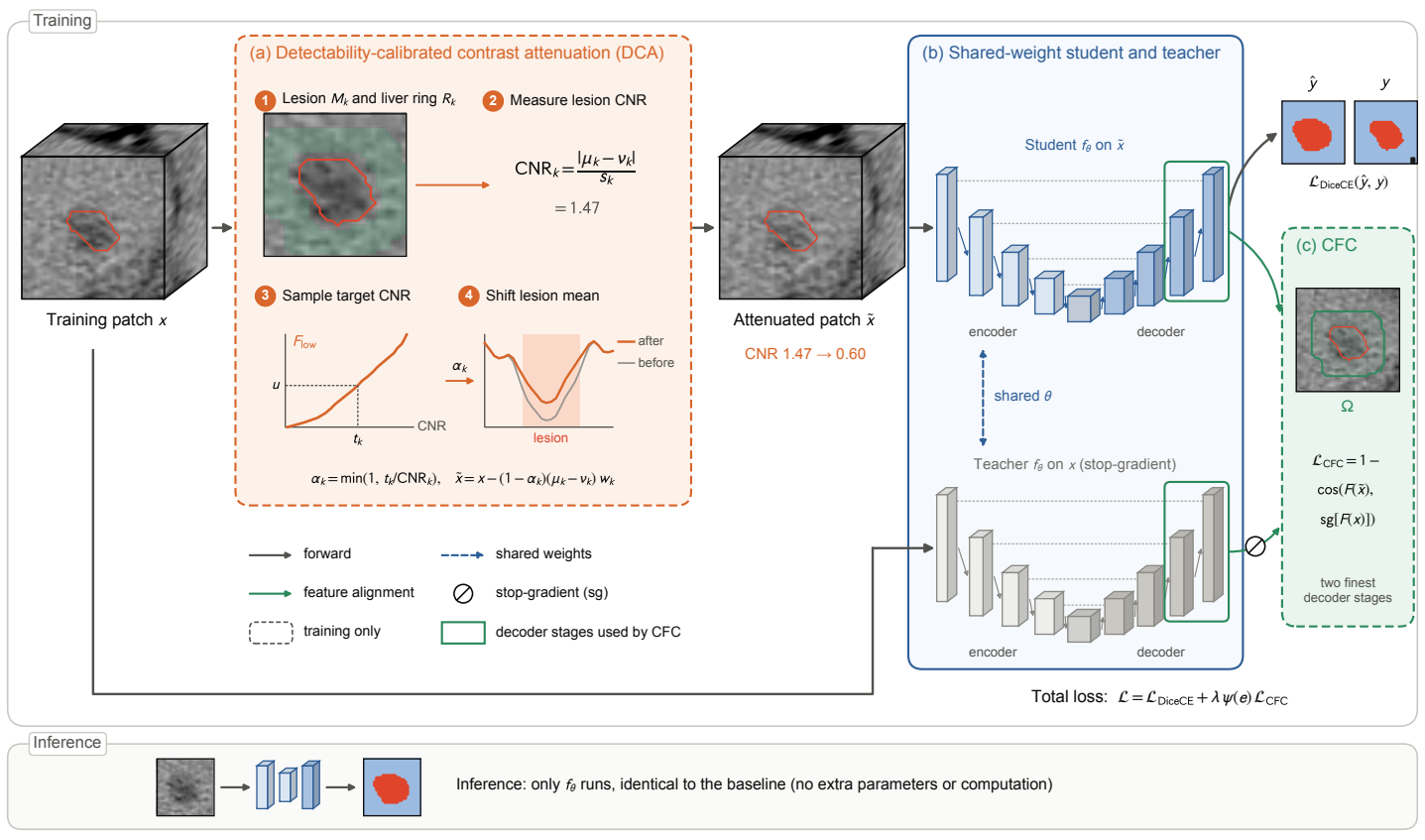

Training
(a) Detectability-calibrated contrast attenuation (DCA)
1 Lesion M_k and liver ring R_k
2 Measure lesion CNR
$\mathrm{CNR}_k = \frac{|\mu_k - \nu_k|}{s_k}$
$= 1.47$
3 Sample target CNR
4 Shift lesion mean
F_{low}
u
t_k
CNR
α_k
after
before
lesion
$\alpha_k = \min(1,\ t_k/\mathrm{CNR}_k), \quad \tilde{x} = x - (1-\alpha_k)(\mu_k - \nu_k)\, w_k$
Training patch x
Attenuated patch $\tilde{x}$
CNR 1.47 → 0.60
(b) Shared-weight student and teacher
Student f_θ on $\tilde{x}$
encoder
decoder
shared θ
Teacher f_θ on x (stop-gradient)
encoder
decoder
$\hat{y}$
y
$\mathcal{L}_{\mathrm{DiceCE}}(\hat{y}, y)$
(c) CFC
Ω
$\mathcal{L}_{\mathrm{CFC}} = 1 - \cos(F(\tilde{x}), \mathrm{sg}[F(x)])$
two finest decoder stages
forward
shared weights
feature alignment
stop-gradient (sg)
training only
decoder stages used by CFC
Total loss: $\mathcal{L} = \mathcal{L}_{\mathrm{DiceCE}} + \lambda\, \psi(e)\, \mathcal{L}_{\mathrm{CFC}}$
Inference
Inference: only f_θ runs, identical to the baseline (no extra parameters or computation)